

CADEMAS-ML: A Software Tool for Context-Aware Cooperative Automated Decision-Making

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Abstract

Automated decision-making systems increasingly operate as part of larger decision ecosystems rather than as isolated models. However, their outputs are often interpreted without considering the organizational, operational, or strategic context in which decisions are made. This paper presents CADEMAs-ML, a software tool that operationalizes the CADEMAs framework for a machine-learning-based class of cooperative and context-aware automated decision-making systems. While CADEMAs is a general conceptual framework for coordinating heterogeneous automated decision units, CADEMAs-ML specializes this idea to settings in which the contributing units are predictive models and the operational goal is case prioritization. The tool assigns weights to predictive components according to selected performance indicators and aggregates their outputs into a global risk score. In parallel, it evaluates contextual suitability using configurable fuzzy membership functions, categorical mappings, derived rules, and aggregation operators. This approach enables decision-makers to represent vague, gradual, and scenario-dependent preferences in a flexible manner. The final prioritization score combines the predictive risk component and the contextual suitability component through an adjustable trade-off parameter. The capabilities of the tool are illustrated through an employee attrition case study that instantiates, rather than redefines, CADEMAs in an ML workflow. Several departmental predictive models are combined with alternative decision contexts, including economic crisis and digital transformation scenarios. The results show that identical predictive risks may lead to different prioritization outcomes depending on the contextual assumptions adopted. CADEMAs-ML therefore provides a practical bridge between cooperative ADM theory, fuzzy context modeling, and auditable decision-support software.

Keywords: automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization

1. Introduction

Automated decision-making (ADM) systems have become an integral part of modern organizations. Following the broad understanding of ADM as computational processes that produce ratings, recommendations, decisions, predictions, or related outputs from data and predefined objectives [1], such systems may include machine-learning classifiers, rule-based procedures, optimization models, simulation components, or other algorithmic decision units. In practice, predictive and classification models are among the most common instantiations, supporting tasks such as credit scoring, fraud detection, medical triage, customer churn analysis, and human-resource management.

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Despite these advances, an important limitation remains. The outputs generated by ADM units, such as recommendations, ratings, class probabilities, or risk scores, are often interpreted in isolation from the organizational environment in which decisions are ultimately made [2,3]. In practice, a technically plausible automated output rarely provides sufficient guidance for action. The appropriate response depends on contextual factors such as organizational priorities, resource constraints, departmental objectives, regulatory constraints, and prevailing strategic conditions. As a result, identical automated assessments may lead to different decisions under different circumstances.

This observation highlights the cooperative nature of real-world decision-making [2]. Decisions are seldom based on a single decision agent or automated component. Organizations frequently rely on heterogeneous units developed for different departments, objectives, data sources, or stakeholder groups. Their outputs must therefore be harmonized into a coherent assessment before they can support action. At the same time, domain experts contribute contextual knowledge that cannot always be represented through precise numerical rules. Such knowledge is often expressed through gradual concepts and linguistic assessments that reflect uncertainty, preference, and strategic intent.

Fuzzy logic provides a well-established mathematical framework for representing this type of contextual reasoning [4,5]. Through fuzzy sets and linguistic variables, experts can model concepts such as urgency, strategic relevance, or suitability in a transparent and interpretable manner. This makes fuzzy approaches particularly attractive for decision-support environments in which quantitative predictions must be complemented by qualitative contextual considerations.

To address these challenges, Novoa-Hernández et al. [6] proposed **CADEMAS** (Cooperative Automated Decision-Making Systems), a conceptual framework for coordinating heterogeneous automated decision units. CADEMAs separates the decision process into an input-management layer, a set of ADM units, an output or decision-integration layer, and a contextual component that evaluates the integrated proposal against external conditions. Importantly, the framework is not restricted to predictive models. A CADEMAs unit may produce a score, recommendation, rule activation, symbolic output, or preliminary decision signal. Predictive probabilities are therefore one possible type of ADM output, not the defining element of the framework.

The practical value of this approach was subsequently illustrated by Godz et al. [7], who instantiated CADEMAs in employee-attrition prevention. Their study did not propose a new CADEMAs variant; rather, it showed how the general architecture can be specialized to a machine-learning setting in which predictive models developed from different organizational perspectives are coordinated with context-dependent management policies to generate prioritizations aligned with organizational objectives.

Although CADEMAs provides a solid conceptual foundation, its adoption is currently limited by the lack of a dedicated software implementation. Researchers and practitioners who wish to apply the framework must still develop their own solutions for routing case data, collecting partial ADM outputs, integrating those outputs, and evaluating the resulting proposal under explicit contextual assumptions. Existing cooperative AI frameworks, multi-agent systems, fuzzy decision-support tools, and hybrid fuzzy-ML applications each address part of this problem, but they do not provide a reusable workflow organized around the CADEMAs separation between decision-unit outputs, integration, context modulation, and final prioritization. As discussed in Section 2.2, this creates a gap between the conceptual framework and its practical application.

This paper addresses that gap by presenting **CADEMAS-ML**, a software tool that operationalizes the CADEMAs framework for cooperative and context-aware ADM. The main contributions of this work are the following:

- Design and implementation of CADEMAs-ML, a software tool that operationalizes CADEMAs for machine-learning-based cooperative and context-aware automated decision-making.
- A configurable methodology that treats predictive models as ADM units and combines their outputs with fuzzy contextual knowledge within a unified prioritization process.
- A transparent and auditable decision-support architecture that exposes both predictive and contextual contributions to the final prioritization score.
- An interactive robustness analysis mechanism that evaluates ranking stability under different balances between predictive evidence and contextual policies.
- Validation through an employee-attribution prevention scenario involving multiple predictive models and alternative organizational contexts.

The remainder of the paper is organized as follows. Section 2 reviews the CADEMAs framework, positions it with respect to adjacent cooperative-decision frameworks, and compares application-oriented decision-support systems related to CADEMAs-ML. Section 3 presents CADEMAs-ML, first describing its architecture and methodological improvements and then illustrating its main functionalities through an employee-attribution case study. Finally, Section 4 concludes and outlines directions for future work.

2. Background and related works

2.1. CADEMAs framework

The CADEMAs (Cooperative Automated Decision-Making Systems) framework was introduced as a general structure for designing systems in which multiple automated decision-making units cooperate to produce contextually aligned decisions [6]. The central motivation behind CADEMAs is not limited to prediction. Rather, it is that real-world decision problems, especially in organizational environments, rarely emerge from a single autonomous unit with a single representation of the problem. Decisions may depend on heterogeneous signals produced by classifiers, optimization models, simulation components, rule-based systems, large-language-model agents, or institutional procedures, each with its own assumptions, data access, output type, and reliability level.

To formalize this idea for the prioritization setting addressed in this paper, let $\mathcal{E} = \{e_1, \dots, e_n\}$ denote a set of cases to be evaluated, and let $\mathcal{A} = \{A_1, \dots, A_K\}$ represent a collection of ADM units. In the general CADEMAs framework, each A_k may produce an output $o_k(e_i)$ whose semantics are not necessarily probabilistic: it may be a class, a score, a ranking, a symbolic recommendation, a rule activation, or another decision object. A decision-integration layer is therefore responsible for transforming these heterogeneous outputs into a shared representation and synthesizing them into an integrated proposal. In the ML specialization implemented by CADEMAs-ML, the ADM units are predictive models and the harmonized output is a scalar risk or priority signal $p_k(e_i) \in [0, 1]$. Since these components may differ in performance and trustworthiness, each A_k is assigned a confidence weight $w_k \geq 0$, with $\sum_{k=1}^K w_k = 1$. These weights can be derived from validation metrics, expert judgment, or governance constraints. The ML specialization aggregates individual outputs into an integrated risk score:

$$R(e_i) = \sum_{k=1}^K w_k p_k(e_i). \quad (1)$$

Rather than treating the integrated output as the final decision criterion, CADEMAs introduces an explicit contextual layer that captures organizational priorities, policies, constraints, and scenario-dependent considerations. The original framework does not prescribe a single context formalism. Context may be represented through crisp rules, utility

functions, regulatory constraints, fuzzy propositions, or other evaluative mechanisms. In CADEMAs-ML, and following the attrition application of Godz et al. [7], fuzzy logic is used because it supports gradual degrees of satisfaction instead of binary rules. Let $\mathcal{C} = \{C_1, \dots, C_L\}$ be a set of contextual factors. For each case e_i , the degree to which it satisfies factor C_ℓ is given by $\mu_\ell(e_i) \in [0, 1]$, and these values are aggregated into a contextual score:

$$C(e_i) = H(\mu_1(e_i), \dots, \mu_L(e_i)), \quad (2)$$

where $H : [0, 1]^L \rightarrow [0, 1]$ is an aggregation operator that can take different forms depending on the decision policy (e.g., t-norms, averages, or hierarchical compositions).

The final prioritization index in CADEMAs-ML is obtained by combining the ML-derived risk and the contextual suitability score:

$$\pi(e_i) = \Psi(R(e_i), C(e_i)), \quad (3)$$

where Ψ is a monotonic function that defines the trade-off between both components. A key design principle inherited from CADEMAs is the separation between the integrated ADM proposal and its contextual interpretation. In the ML specialization, this makes explicit what the predictive models suggest and what the organization considers relevant.

The framework has been instantiated in employee attrition prevention by Godz et al. [7]. In that application, multiple machine-learning models were trained on different organizational perspectives (e.g., Human Resources, Finance, Operations, and Employee Wellbeing). Each model acts as an ADM unit within $\mathcal{A}$, and their outputs are combined using weights derived from validation performance, specifically AUC. In parallel, the contextual layer encodes scenario-dependent organizational priorities (e.g., economic crisis or digital transformation settings) using fuzzy membership functions and rule-based structures. This work should therefore be read as an application of CADEMAs to an ML-driven decision-support problem, not as a reformulation of the general CADEMAs framework.

Although this implementation demonstrates the practicality of the framework, it represents only one possible configuration. In general, CADEMAs is designed to remain flexible across several dimensions: the type of ADM units, the semantics of their outputs, the mechanism used for semantic harmonization, the integration operator, and the context-modulation strategy. CADEMAs-ML fixes some of those degrees of freedom in order to provide a concrete software tool: ADM units are H2O MOJO predictive models, the integrated output is a weighted probability-based risk score, the contextual model is expressed through fuzzy rules, and the final decision is a tunable prioritization index. Within that specialization, confidence weights w_k can still be defined using alternative performance measures (e.g., F1-score, precision, recall, calibration metrics, or utility-based indicators), the aggregation operator H may reflect different contextual policies, and the combination function Ψ can be tuned to represent different trade-offs between risk and context.

This flexibility is essential for adapting CADEMAs to heterogeneous organizational environments and motivates the need for a software implementation that operationalizes these design choices in a systematic and configurable way.

2.2. Related frameworks and decision-support applications

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address

how multiple decision entities interact, and application-oriented decision-support systems, which implement concrete methods for a domain or family of decision problems.

At the framework level, Tariq et al. [2] propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Similarly, Varga et al. [3] discuss cooperative AI architectures for high-stakes environments and argue that effective collaboration requires a shared representation of context between automated models and domain experts. Other recent systems show related structural patterns in more specific communities: multi-agent LLM planning [8], cooperative driving with LLM-based modules [9], trust-aware multi-agent reinforcement learning [10], distributed decision coordination [11], and ethical multi-agent governance [12]. These works are important alternatives or neighboring frameworks because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION [13] and the fuzzy DEMATEL-based DSS described by Chekry et al. [14] emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored the combination of machine learning and fuzzy reasoning. The survey by Kumar et al. [4] highlights the growing interest in hybrid fuzzy-ML systems and identifies configurability and expert participation as key challenges for future decision-support tools. Other studies have proposed application-specific integrations of predictive models and expert knowledge, including fuzzy decision-support architectures [5], Fermatean fuzzy decision frameworks [15], and context-sensitive prioritization approaches based on fuzzy AHP and machine learning [16]. These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs decomposition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including PYFDM [17], PYIFDM [18], FuzzyR [19], the intuitionistic-fuzzy toolkit of Roszkowska and Wachowicz [20], the ELIGERE platform [21], and the Fuzzy Cognitive Maps package presented by Dikopoulou and Papageorgiou [22], are also relevant. They improve reproducibility and provide reusable computational building blocks. Nevertheless, software availability is not the central distinction pursued here. The key issue is whether a system captures the cooperative ADM problem that CADEMAs is intended to solve: multiple decision-producing units, explicit integration of their outputs, contextual evaluation that is kept conceptually separate, and a final decision mechanism whose assumptions can be inspected and varied.

Table 1 summarizes representative application-oriented contributions in these areas. Two observations emerge from this comparison. First, several works provide valuable fuzzy, MCDM, or hybrid fuzzy-ML mechanisms for transparent decision support. Second, these mechanisms usually address one part of the CADEMAs problem: preference aggregation, explainable fuzzy rules, contextual scoring, or predictive assistance. What remains missing is an operational workflow in which multiple ADM units are treated as cooperating components, their outputs are integrated explicitly, contextual evaluation is modeled as a separate layer, and the final prioritization can be recomputed under different decision policies.

CADEMAs-ML is designed to address this gap for the predictive-model setting. Building upon the conceptual framework introduced by Novoa-Hernández et al. [6] and the employee-attribution instantiation reported by Godz et al. [7], it provides a configurable soft-

Table 1. Comparison of representative application-oriented decision-support works related to CADEMAs-ML. The comparison emphasizes whether each contribution captures the cooperative ADM elements targeted by CADEMAs, rather than whether software is publicly released.

Work	Decision-support focus	Multiple ADM units	Explicit context layer	Relation to CADEMAs-ML
Godz et al. [7]	Employee-attribution prevention using CADEMAs with departmental ML models	Yes	Yes	Direct application that motivates the implemented workflow
Więckowski and Sałabun [13]	Auditable fuzzy MCDM through a web-based tool	No	Partly	Strong transparency, but no cooperative ADM integration layer
Chekry et al. [14]	Fuzzy DEMATEL decision-support system	No	Partly	Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like
Ghosh et al. [5]	Fuzzy DSS with integrated expert rules for predictive workflows	Partly	Yes	Combines prediction and expert rules, but as a bespoke workflow
Aly et al. [16]	Context-sensitive prioritization using fuzzy AHP and ML	Partly	Yes	Hybrid fuzzy-ML prioritization without a general ADM architecture
Suvitha et al. [15]	Fermatean fuzzy aggregation for expert-driven preferences	No	Partly	Rich aggregation operators, but no predictive/ADM cooperation layer
CADEMAs-ML (this work)	Configurable prioritization with ML-based ADM units and fuzzy context scenarios	Yes	Yes	Operationalizes CADEMAs for reusable ML-driven decision support

ware implementation in which ML-based ADM outputs and contextual reasoning coexist as explicit and auditable components of the decision-making process. Its planned open-source availability is an additional benefit, but the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

3. CADEMAs-ML software tool

This section presents CADEMAs-ML as a software realization of one concrete CADEMAs specialization: cooperative prioritization supported by multiple predictive ADM units and an explicit fuzzy context layer. We first explain how the general CADEMAs architecture is instantiated in a configurable ML workflow. Special attention is given to the fuzzy context model and to the robustness analysis of the prioritization ranking. We then use the employee-attribution example distributed with the repository to show the main functionalities of the application.

3.1. Architecture and improvements

CADEMAs-ML is not a domain-specific attrition script. It is an operational implementation of CADEMAs for the case in which the cooperating ADM units are predictive models. In line with Novoa-Hernández et al. [6], the software keeps partial automated assessments separate from their contextual interpretation. Its contribution is to make this

Figure placeholder: Architecture of CADEMAs-ML showing the four inputs, MOJO inference, fuzzy context evaluation, λ -controlled prioritization, and the output modules.

Figure 1. Operational architecture of CADEMAs-ML. The software instantiates the CADEMAs architecture as a configurable pipeline in which predictive risk and contextual alignment are computed independently before being integrated into a prioritization index.

separation executable for ML-based decision support. Predictive components, contextual rules, and decision policies are configured explicitly and can be inspected through the interface. Thus, the attrition study of Godz et al. [7] becomes one possible configuration of the tool, not its only intended use.

The application combines a Streamlit interface, H2O MOJO inference, and a vectorized fuzzy-context engine. A run starts from four inputs: the case dataset, the MOJO models, a model-metadata JSON file, and one or more context JSON files. These inputs map naturally onto the CADEMAs layers. The dataset and configuration files act as the input manager. The MOJO files instantiate the ADM units in this ML specialization. Their weighted probabilistic outputs form the integrated predictive risk. The fuzzy-rule engine provides the contextual model. The final layer combines both streams through a user-adjustable policy parameter. Figure 1 summarizes this architecture.

3.1.1. ML-based cooperative risk

The predictive layer follows the cooperative aggregation principle of CADEMAs as specialized in the attrition instantiation of Godz et al. [7]. CADEMAs-ML exposes that specialization as a reusable software component. Let $\mathcal{E} = \{e_1, \dots, e_n\}$ be the set of cases and let $\mathcal{M} = \{M_1, \dots, M_K\}$ be the uploaded predictive ADM units. Each model M_k produces, for case e_i , a positive-class probability

$$\hat{p}_{ik} \in [0, 1].$$

The models need not use the same feature subset. Each MOJO receives the available case matrix and uses its internal schema to select the required variables. The contribution of each component is controlled by a non-negative confidence weight w_k . This weight is computed from a user-selected validation metric s_k :

$$w_k = \frac{s_k}{\sum_{\ell=1}^K s_{\ell}}, \quad \sum_{k=1}^K w_k = 1. \quad (4)$$

The cooperative machine-learning risk score is then

$$R_i = \sum_{k=1}^K w_k \hat{p}_{ik}. \quad (5)$$

This mathematical form is the ML-specific realization of the more general CADEMAs decision-integration layer. The improvement introduced by CADEMAs-ML is practical configurability. Users can upload any number of compatible MOJO models, read model metadata from JSON, and choose the validation metric used for weighting.

3.1.2. Configurable fuzzy context model

The context layer is where CADEMAs-ML most clearly turns the conceptual framework into an operational artifact. The role of context follows Novoa-Hernández et al. [6], and the use of fuzzy propositions follows the attrition application in Godz et al. [7]. The difference is that CADEMAs-ML does not hard-code those contexts. It represents them as JSON specifications. The same formalism can therefore be reused with other domains, rule families, and decision policies.

Formally, a context is defined as a collection of fuzzy propositions. Let V_r be a case attribute and let $\tilde{F}_r$ be a fuzzy set representing a linguistic predicate over the universe of V_r . For case e_i , an atomic rule r evaluates the proposition “ V_r is $\tilde{F}_r$ ” as

$$\mu_r(e_i) = \mu_{\tilde{F}_r}(x_{ir}) \in [0, 1], \quad (6)$$

where x_{ir} is the observed value of attribute V_r for case e_i . The implementation supports triangular, trapezoidal, linear increasing, and linear decreasing membership functions. It also supports categorical maps and categorical sets. The former assign degrees directly to symbolic values. The latter group categories into linguistic labels that can be scored later in derived rules.

Atomic rules may include activation conditions. If $m_r(e_i) \in \{0, 1\}$ denotes whether the when condition of rule r is satisfied for case e_i , the effective membership is

$$\mu_r^*(e_i) = m_r(e_i) \mu_r(e_i), \quad (7)$$

with $m_r(e_i) = 1$ for unconditional rules. This makes it possible to represent policies such as department-specific salary suitability without duplicating datasets or writing procedural code.

Beyond atomic rules, CADEMAs-ML allows the definition of derived rules. A derived rule d combines a subset of already computed memberships through an aggregation operator G_d :

$$\mu_d(e_i) = G_d(\mu_{r_1}^*(e_i), \dots, \mu_{r_q}^*(e_i)). \quad (8)$$

The implemented operators are minimum-based AND, maximum-based OR, algebraic PRODUCT, and arithmetic AVERAGE. The context alignment score is then obtained by applying a final aggregation policy H to selected atomic and derived memberships:

$$C_i = H(\mu_1(e_i), \dots, \mu_L(e_i)). \quad (9)$$

The function H can be selected interactively or specified as a declarative logic tree in the context JSON. This captures the attrition instantiation in Godz et al. [7], but makes it portable. It also lets organizations express conservative, compensatory, hierarchical, or conditional contextual policies without changing the application code.

All atomic and derived memberships are retained as an audit matrix. The final context score is therefore not a black-box modifier of the ML risk. It can be decomposed into the rule activations that produced it.

3.1.3. Decision integration and robustness

Predictive evidence and contextual suitability remain separate until the final decision stage. For a fixed policy parameter $\lambda \in [0, 1]$, CADEMAs-ML computes the prioritization score as

$$P_i(\lambda) = \lambda R_i + (1 - \lambda) C_i. \quad (10)$$

The interpretation is direct. A value $\lambda = 1$ produces a risk-driven ranking. A value $\lambda = 0$ produces a context-driven ranking. Intermediate values encode explicit trade-offs between both components.

The robustness module is a specific improvement of CADEMAs-ML over the previous static attrition instantiation. Let $\Lambda = \{\lambda_0, \dots, \lambda_q\}$ be a finite grid of policy values. For each $\lambda \in \Lambda$, the tool computes the score vector $P(\lambda)$ and the ranking

$$r_i(\lambda) = 1 + |\{j : P_j(\lambda) > P_i(\lambda)\}|, \quad (11)$$

with deterministic tie-breaking by case order. This produces a rank trajectory for every case. CADEMAs-ML summarizes these trajectories through a bump chart, a rank box plot, and rank acceptability indices. The bump chart shows how each case moves as λ changes. The box plot summarizes the distribution of ranks over Λ . Cases are ordered by median rank and interquartile range. Finally, the rank acceptability index estimates how often a case attains a given rank:

$$b_i^r = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} \mathbf{1}\{r_i(\lambda) = r\}. \quad (12)$$

Together, these indicators distinguish stable priorities from policy-sensitive cases.

3.2. Main functionalities

We now describe the main functionalities of CADEMAs-ML through the employee-attrition example distributed with the application. The example follows the rationale of Godz et al. [7], but is presented here as a reproducible software workflow. The goal is not to improve attrition prediction accuracy. Rather, it is to show how heterogeneous predictive assessments and contextual policies can be configured, audited, and combined for intervention prioritization.

3.2.1. Input configuration and analysis workflow

The case study is based on the IBM HR Analytics Employee Attrition and Performance dataset used by Godz et al. [7]. The original data contain 1,470 employee records and 35 attributes describing demographic, professional, salary, performance, and organizational conditions. The repository includes a 20-case subset prepared to demonstrate the CADEMAs-ML workflow. The subset is smaller in records, not in feature coverage: after excluding the `Case_ID` identifier, it keeps the same 35 dataset columns, including the known attrition label (10 employees labeled `Yes` and 10 labeled `No`). This label is useful for verification when inspecting the results, but it is not used as a prioritization criterion. Appendix A lists the 34 predictor variables and the subset used by each MOJO model.

The example bundle contains a semicolon-delimited CSV file, a model-definition JSON file, two context JSON files, and four H2O MOJO models. The CSV file has a `Case_ID` column with employee names. CADEMAs-ML uses this column as the case identifier. If no

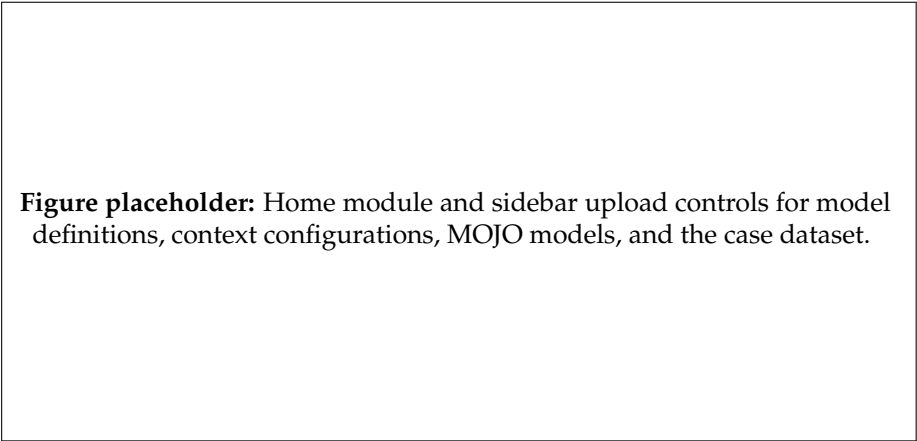


Figure 2. Home and input configuration module. CADEMÁS-ML receives four inputs and presents the operational flow from data loading to prioritization and robustness analysis.

Table 2. Models used in the attrition example and AUC-based weights computed by CADEMÁS-ML.

Component	Organizational perspective	AUC	Weight
M_1	Culture and Wellbeing	0.9941	0.2623
M_2	Finance	0.9972	0.2631
M_3	Operations and Development	0.8022	0.2116
M_4	Human Resources	0.9968	0.2630

identifier is provided, the application creates one automatically. Identifier and outcome columns are excluded from model inference.

Following the training protocol of Godz et al. [7], the predictive layer simulates four organizational perspectives: Culture and Wellbeing, Finance, Operations and Development, and Human Resources. Each perspective is represented by an H2O AutoML model exported as a MOJO file and reads a distinct feature subset from the shared case matrix (Table A1). Using AUC as the weighting metric, the reproduced run produced the weights shown in Table 2. The Operations and Development component receives a lower weight because its validation AUC is lower than that of the other three models.

3.2.2. Overview: prioritized cases

After execution, the Overview module presents global indicators, a risk-context scatter plot, a score distribution, and the prioritized cases table. In the reproduced Economic Crisis scenario with $\lambda = 0.5$, the ML risk ranged from 0.0012 to 0.9882. The context score ranged from 0.1385 to 0.6892. The final prioritization score ranged from 0.1528 to 0.7401. Table 3 reports the top eight cases.

The top cases illustrate the logic of the system. The first-ranked employee does not have the highest ML risk alone. Rather, the case combines very high risk with moderate contextual alignment under the economic-crisis policy. Conversely, a case with strong contextual alignment may lose priority if its predicted attrition risk is low. This is why CADEMÁS-ML keeps R_i and C_i separate before applying the policy trade-off.

3.2.3. Models: inspecting cooperative risk

The Models module exposes the ML side of the computation. It displays the normalized model weights, the individual probabilities produced by each MOJO, and the aggregated score R_i . This makes the cooperative risk auditable. A high value of R_i can be traced back to one model, to several aligned models, or to the influence of highly weighted components.

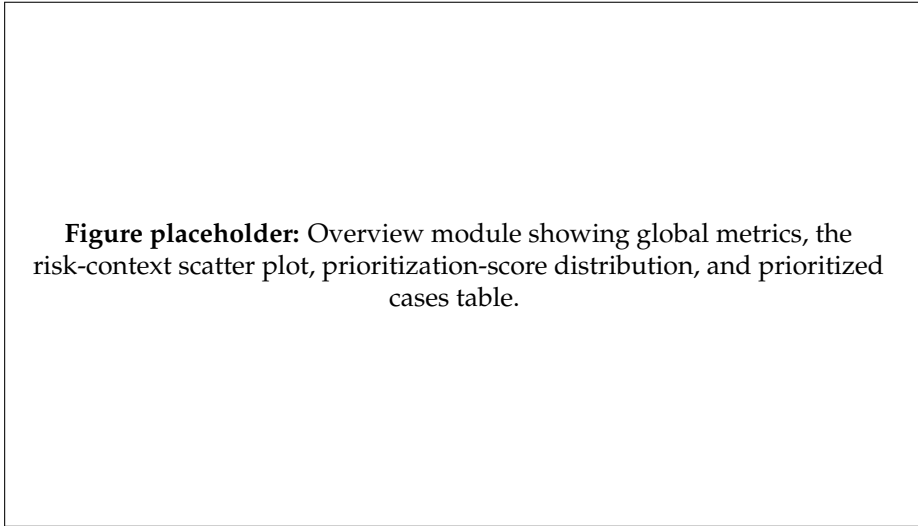


Figure 3. Overview module for the attrition example. The module summarizes the global behavior of the run and identifies cases with the highest prioritization scores.

Table 3. Top prioritized employees in the Economic Crisis context for $\lambda = 0.5$.

Case	R_i	C_i	$P_i(0.5)$	Rank
Theodore Adams	0.9803	0.5000	0.7401	1
Amelia Davis	0.9879	0.4895	0.7387	2
Evelyn Taylor	0.9555	0.4542	0.7048	3
William Young	0.9751	0.4098	0.6925	4
Charlotte Miller	0.9204	0.3906	0.6555	5
Henry King	0.9676	0.3308	0.6492	6
Abigail Anderson	0.9796	0.2923	0.6359	7
Benjamin Scott	0.9622	0.2923	0.6272	8

3.2.4. Context: auditing fuzzy policies

The Context module presents the fuzzy-policy side of the computation. In the Economic Crisis scenario, the policy prioritizes employees who are hard to replace, have relevant tenure, show at least medium performance, have acceptable salary cost within their department, and display medium or high commitment. The context is encoded with 11 atomic rules and three derived rules: `hard_to_replace`, `commitment_medium_or_high`, and `salary_cost_ok`. In the reproduced run, C_i had mean 0.4057 and maximum 0.6892. The highest context-aligned cases were Oliver Walker, Noah Lewis, Sophia Brown, Theodore Adams, and Amelia Davis.

The second context, Digital / Technological Transformation, focuses on upskilling capacity, role adaptability, career scalability, and investment horizon. It uses seven atomic rules and three derived rules. Its context scores ranged from 0.1500 to 0.7100. The top context-aligned cases were Noah Lewis, Evelyn Taylor, Oliver Walker, Emma Smith, and Olivia Johnson. The contrast between both contexts shows that CADEMAs-ML is not tied to the policy choices in Godz et al. [7]. A different organizational situation can be represented by changing the JSON context specification.

3.2.5. Robustness: sensitivity of the priority ranking

The Robustness module evaluates whether the prioritization is stable when λ changes. For the Economic Crisis run, we used the grid $\Lambda = \{0, 0.25, 0.5, 0.75, 1\}$. Amelia Davis was highly stable among the top alternatives, with median rank 2 and interquartile range 0. Theodore Adams also remained near the top, with median rank 2 and rank range 1–4. Lucas Wright behaved differently. This case ranged from rank 1 to rank 19, which indicates

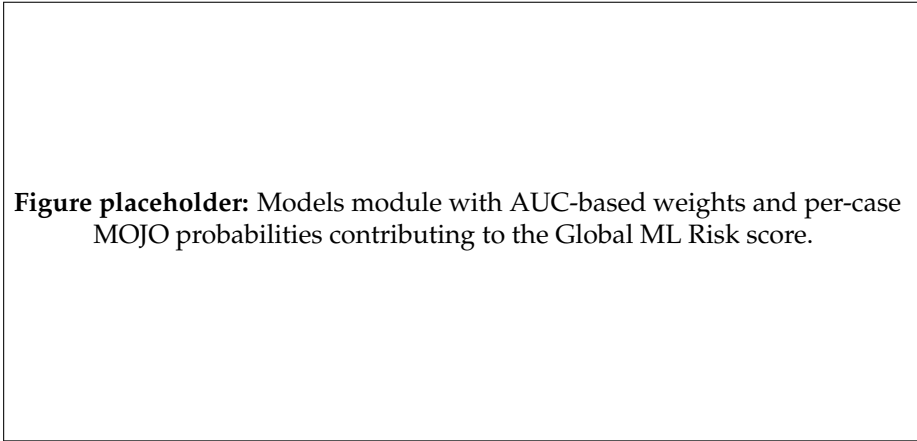


Figure 4. Models module. The table links each uploaded MOJO to its confidence weight and displays the per-case probability outputs used to compute R_i .

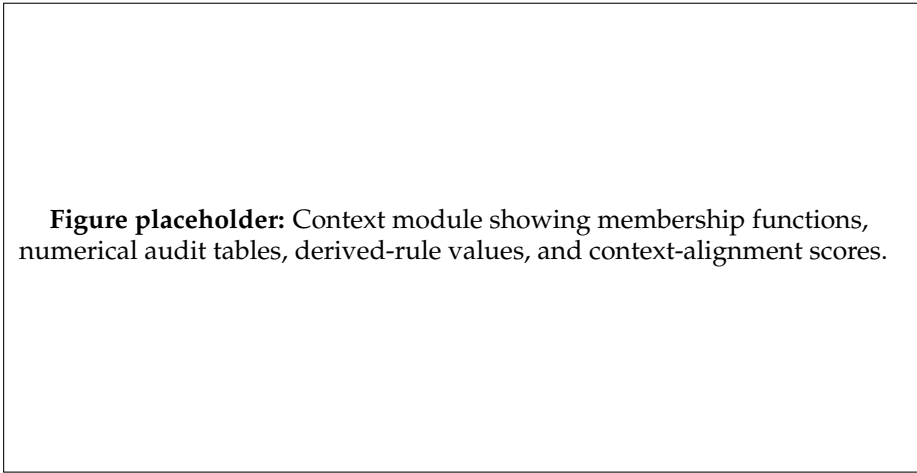


Figure 5. Context module. Atomic and derived membership values are exposed for audit, allowing analysts to inspect why a case receives a given contextual score.

strong sensitivity to the policy trade-off. It is favored under context-driven evaluation, but loses priority as the ranking becomes risk-driven.

The rank acceptability index complements the bump chart by estimating how often each case attains each rank over the λ grid. For example, Amelia Davis occupied rank 2 in 60% of the evaluated policy settings. Benjamin Scott occupied rank 8 in 60% of them. These summaries add an interpretable robustness layer. They help distinguish cases that are consistently high priority from those whose position depends strongly on the chosen balance between predictive risk and contextual policy.

4. Conclusion and future works

Appendix A. Attrition example dataset and model features

The 20-case attrition example distributed with CADEMAs-ML contains 36 columns: Case_ID, 34 predictor variables, and the attrition outcome label. The predictors cover demographic, professional, compensation, performance, and organizational attributes from the IBM HR dataset used by Godz et al. [7]. Five variables (age, gender, department, job_role, and job_level) are shared by all four MOJO models. Two additional columns (employee_count and employee_number) are retained in the CSV file but are not used by any model in the example bundle. Each MOJO receives the full case matrix and selects its required variables according to the schema stored in the model-definition JSON file.

Figure placeholder: Robustness module with bump chart, rank box plot, and Rank Acceptability Indices heatmap.

Figure 6. Robustness module. The module evaluates rank stability under changes in λ , making visible both robust priorities and policy-sensitive cases.

Table [A1](#) summarizes the feature assignment. The Human Resources model reads
over18 from the CSV file; the corresponding field is named over_18 in the model metadata.

Table A1. Predictor variables in the attrition example and the MOJO models that use each one. M_1 – M_4 denote Culture and Wellbeing, Finance, Operations and Development, and Human Resources, respectively.

Predictor	M_1	M_2	M_3	M_4
age	✓	✓	✓	✓
business_travel	✓			
daily_rate		✓		
department	✓	✓	✓	✓
distance_from_home				✓
education				✓
education_field				✓
employee_count				
employee_number				
environment_satisfaction	✓			
gender	✓	✓	✓	✓
hourly_rate		✓		
job_involvement	✓			
job_level	✓	✓	✓	✓
job_role	✓	✓	✓	✓
job_satisfaction	✓			
marital_status				✓
monthly_income		✓		
monthly_rate		✓		
num_companies_worked			✓	
over18				✓
over_time	✓			
percent_salary_hike		✓		
performance_rating			✓	
relationship_satisfaction	✓			
standard_hours				✓
stock_option_level		✓		
total_working_years			✓	
training_times_last_year			✓	
work_life_balance	✓			
years_at_company			✓	
years_in_current_role			✓	
years_since_last_promotion			✓	
years_with_curr_manager			✓	

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